Computer Networks
Assignment -3

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Date:

Review Questions

R3. A → B Sp x dp y
B → A Source port number y
Destination port number x

R4. He may not choose to use TCP's congestion control, which adds significant overhead. UDP is better option as it does not use congestion control while it does not gurantee reliable data transfer ~~prot~~ of TCP.

R5 Most internet firewalls block UDP traffic, so TCP is used instead. Data of voice and video also has to be transmitted reliably by TCP.

R11 If round trip time is known then sender knows if either packet or ACK/NACK for packet is lost compared to real scenario where ACK/NACK might be on the way to sender after timer expires
TO detect loss for each packet a timer of constant duration will be necessary for sender.

Problems

P40: a) TCP slow start 1 -> 6
and 23 -> 26

b) TCP congestion avoidance 6 -> 16
and 17 -> 22

c) After 16th transmission round
Packet loss is recognized by
triple duplicate ACK
In case of time out congestion window
would drop to 1

d) After 22nd transmission round
segment loss is detected due to timeout
, the congestion window size is set to 0.

ssthresh e) the threshold is initially 32
at that point slow start stops and
congestion avoidance begins

f) the threshold is set to half of congestion
window when next packet loss is detected.
So when loss is detected at transmission round 16
Congestion size is 42, hence threshold is 21 at
round 18.

g) when loss is detected at transmission round 22
congestion size is 29, hence threshold will be
14 in 24th transmission round.

Date:

	transmission round	packet
h)	1	1
	2	2-3
	3	4-7
	4	8-15
	5	16-31
	6	32-63
hence at 7th transmission round	7	64-96

packet 70 is sent.

i) threshold will be half of congestion window (8) when loss is detected, congestion window will be set to the new threshold + 3 Max segment size the threshold is 4 cwnd / window will be 7

j) threshold → 21 congestion window → 1

	round	packet
k)	17	1
	18	2
	19	4
	20	8
	21	16
	22	21 +
		52 → 52 packets

Question 2

When host A sends seq N = 500, and bytes to send are 46 hence next ~~[illegible]~~ seq N from host B is <u>546</u>

Question 3:

UDP checksum -> sum the 8 bit words

```
  1111  1010  -> 050
+ 0110  0101  -> 101
  0101  1111  -> 351 -> 9 bits
1
```

```
sum      0110  0000
checksum
(1's compl)  1001  1111
```

reciever compute checksum

```
  1111  1010
+ 0110  0101
  1001  1111
  1111  1111   all 1s
```

Question 4/

a) 20 bytes in first segment

Second seg starts at 110 so first seg ~~90-109~~ -> 20 bytes
110-90

b) TCP uses cumm ACKs so last successful ACK 90 will be sent which is Ack 90